

WFP INNOVATION ACCELERATOR x AFCIA LAC 2026

AMARU

Anticipatory Climate Intelligence for Food Security

Bi-Climatic Early Warning: El Niño (Floods) & La Niña (Frosts / Droughts)

Seed Stage (TRL 6) | Peru

Founder & Project Lead: Carlos Baños Díaz (MBA in Business Administration)

Corporate Shared Services Experience (tgestionona) | Executive Diplomas: Public Governance (PUCP/CAF) & Public Sector AI (U. del Pacífico)

1. The Problem: The Double Climate Threat in Peru

Severe impacts on smallholder farming, high-altitude pastoralism, and food supply chains

■ El Niño Costero: Flash Floods & Debris Flows

- **Irrigation Collapse:** Destruction of rustic community water intakes (bocatomas) and main canals.
- **Crop Washout:** Flooding of thousands of hectares of family subsistence and cash crops.
- **Logistical Severance:** Highway blockages rotting perishable harvests in transit to urban markets.
- **Lives at Risk:** Sudden debris flows (huaicos) hitting inhabited alluvial fans with zero lead time.

❄️ ■ La Niña ('Chiri'): Extreme Frosts & Cold Snaps

- **Mass Livestock Mortality:** Hypothermia and forage loss killing alpacas, sheep, and cattle.
- **Crop Destruction:** Severe agronomic freezes annihilating potato, corn, and quinoa harvests.
- **Acute Food Insecurity:** Subsistence farming families lose an entire year of sustenance.
- **High Vulnerability:** Remote Andean basins lack micro-climatic frost forecast resolution.

■ ■ **Public Sector Bottleneck:** Reactive disaster governance (post-disaster EDAN forms). Subnational municipalities chronically under-execute allocated prevention funds (PP0068 / FONDES) due to a complete lack of objective, auditable technical evidence prior to disaster occurrence.

2. Technological Innovation: Multi-Agent Indicators & Governance

Deterministic multi-agent architecture powered by official micro- and meso-scale physical indicators

■ Bi-Climatic Indicators

- **IPH-FEN:** Debris Flow Forecast Index (slope stability, Manning flow velocity, and bedload shear stress).
- **ISH-CHIRI:** Frost Severity Index (Stefan-Boltzmann nocturnal radiation balance and wind-chill).
- **IRCE-FEN:** District-Level Critical Risk Index ($\text{Hazard} \times \text{Vulnerability} \div \text{Response Capacity}$).

■ Sovereign AI Governance

- **Peru AI Law N° 31814:** Strict human-in-the-loop oversight; no autonomous budget or evacuation triggers.
- **Open Physics Models:** Zero dependency on unexplainable black-box third-party algorithms.
- **Explainability:** Auditable physical equations designed for technical decision support.

■ Cryptographic Traceability

- **SHA-256 Hashing:** Every pericial bulletin generates an immutable, timestamped cryptographic proof.
- **Legal Audit Trail:** Provides tamper-proof technical backing for local authority emergency auditing.
- **1,891 Districts:** Master national catalog harmonized under Supreme Decree N° 234-2025-EF.

3. Value Proposition: Anticipatory Design Hypotheses

Technical lead-time targets formulated as design hypotheses to be calibrated empirically in the field

■ IPH-FEN & IRCE-FEN (El Niño)

- **Design Target:** Up to 72h lead time to close canal intake gates and protect irrigation assets.
- **Kinetic T_lag:** Kirpich calculation of floodwave arrival time from mountain crest to intake.
- **Calibration Goal:** Verified against real-time physical river gauge levels and ANA telemetry.

❄️ ■ ISH-CHIRI (La Niña)

- **Design Target:** Up to 48h advance warning prior to acute sub-zero temperature drops.
- **Cryoclimate:** Inversion layer monitoring and nocturnal temperature gradient tracking.
- **Actionable Mitigation:** Moving alpaca/sheep herds to shelters and distributing dry forage.

■ Autonomous Road Logistics

- **AMARU Route Intelligence:** Proprietary projection of road cuts based on debris flow and bridge scour risk.
- **Outperforming State Lag:** Overcomes the multi-day reporting delay of official highway viewers.
- **Food Preservation:** Identifying safe transit corridors to evacuate perishable crops in time.

Progressive Staging: During the 9-month Sprint, we will validate and calibrate these hypotheses in priority agricultural and pastoral pilot basins, with the software architecture built to expand nationwide.

4. Current Maturity & Evidence (TRL 6 - Seed Stage)

Market-ready software platform validated through automated testing, historical data, and live bots

■ Functional Software Architecture

- **Multi-Agent Engine:** Operational multi-agent core integrating hydrological, georisk, and legal agents.
- **Automated Test Suite:** 167 unit tests passing with pytest (100% test coverage).
- **C2 Command Dashboard:** Functional command-and-control user interface displaying district alert levels.
- **Live Alert Dispatch:** Automated bot operational and actively dispatching alerts via Telegram.

■ Calibrated Indicators & Clocks

- **Core Indicators:** Initial calibration of **IPH-FEN** (debris flows), **IRCE-FEN** (districts), and **ISH-CHIRI** (frosts) benchmarked with SENAMHI/ANA series.
- **Tactical Polar Clocks:** El Niño Clock (November Golden Window) and La Niña Clock (April Golden Window for livestock vaccination/PREVAED).
- **Historical Backtesting:** Validated against historical mega-events: 1983, 1998, 2017, and Cyclone Yaku 2023.

5. Competitive Advantage & Differentiation

Why AMARU outperforms conventional static maps and outdated government viewers

State Viewers (CENEPRED / SAECOE)

- Static retrospective GIS based on historical PDFs.
- No real-time hydrodynamic river routing.
- SAECOE road closures reported days after the cut.
- No direct automated mobile alerts to farmers.

Global Weather APIs

- Generic atmospheric summaries (mm rain or °C).
- Blind to local Peruvian riverbeds and steep slopes.
- Cannot model irrigation canal intake vulnerability.
- No alignment with public finance budget execution.

AMARU (FEN & CHIRI)

- **Dynamic Physical Modeling:** Calculates IPH-FEN (slopes), IRCE-FEN (districts), and ISH-CHIRI (frosts).
- **Direct Push Alerts:** Active Telegram bot, expanding to multi-carrier SMS and VHF radio.
- **Autonomous Route Intelligence:** Anticipatory transport hazard projection.
- **Public Budget Backing:** Prepared (subject to validation) to justify PP0068/FONDES spending.

6. Pilot Implementation Plan (9-Month Sprint)

Field calibration, community validation, and deployment pathway supported by WFP

Months 1 - 3: Setup

- **WFP Bootcamp:** Alignment with WFP innovation teams in Munich / virtual.
- **Field Partnerships:** Formalizing agreements with pilot Irrigation Boards and pastoral co-ops.
- **Technical Onboarding:** Contracting Lead Hydraulic Advisor and Field Agronomist.
- **IoT Deployment:** Mounting 4 non-invasive ultrasonic nodes (per internal Technical Sheet).

Months 4 - 6: Calibration

- **Indicator Tuning:** Calibrating Manning coefficients and frost thresholds with telemetry and SENAMHI.
- **Omnichannel Expansion:** Deploying multi-network SMS gateway and community VHF radio.
- **Farmer Drills:** Simulating early warning and gate-closing protocols with irrigation leaders.

Months 7 - 9: Validation

- **Active Season:** Live operational testing during seasonal rain and frost events.
- **Impact Auditing:** Measuring avoided crop and livestock losses with community members.
- **Pericial Certification:** Auditing technical reports with municipal civil defense officers.
- **WFP Demo Day:** Presenting validated results and regional scaling roadmap.

7. Team & Talent Onboarding Plan (USD 28,000 Budgeted)

Experienced corporate leadership backed by specialized engineering hires funded by the grant

■ Current Project Leadership

- **Founder & Project Lead:** Carlos Baños Díaz (99% equity ownership, full-time dedication).
- **Business Leadership:** MBA in Business Administration.
- **Corporate Operations Background:** Prior corporate experience at tgestiona managing Shared Services Centers (BPO/SSC), bringing deep expertise in operational scalability.
- **Public Sector & AI Diplomas (in progress):** Public Governance & Innovation (PUCP/CAF) & Strategic AI in Public Sector (U. del Pacífico).
- **Technical Role:** Software architect and creator of the AMARU multi-agent platform.

■ Key Roles to be Funded by the Grant (USD 28,000)

- **Lead Hydraulic Engineering Advisor (Part-Time):** Senior water resources specialist (UNI / CIP engineering network) for physical hydrodynamic model co-validation.
- **Local Field Agronomist (Full-Time):** Agricultural engineer based in the pilot basin for direct liaison with Irrigation Boards and pastoralist communities.
- **IoT Telemetry Specialist (Deliverable-Based):** Field technician for mounting ultrasonic sensors and maintaining low-bandwidth gateways.

8. Social Impact, Livestock & Last-Mile Inclusion

Engineered to protect vulnerable rural smallholders without high-tech connectivity barriers

■ Rural Food Security

- **Smallholder Agriculture:** Safeguards the livelihoods of family farmers reliant on canal irrigation.
- **High-Altitude Herds:** Anticipates acute frosts (ISH-CHIRI) to shelter vulnerable alpacas and sheep.
- **Saving Human Lives:** Life-saving evacuation lead time before debris flows strike inhabited zones.

■ Smartphone-Free Inclusion

- **140-Byte Payloads:** Ultra-light compressed alerts compatible with standard SMS on basic 2G phones.
- **Community VHF Radios:** Automated audio bulletins for remote Andean zones lacking cellular towers.
- **Indigenous Languages:** Strategic roadmap for alert dissemination in native Quechua.

■ Environmental Safeguards

- **WFP Category C:** Negligible or zero adverse environmental and social risk.
- **Non-Invasive Telemetry:** Solar-powered sensors mounted on bridge decks without in-stream disruption.
- **Zero Displacement:** Pure anticipatory digital intelligence involving no expropriation.

9. Financial Sustainability & Vision with WFP

Prudent financial projections and national scaling across 1,891 districts to #DisruptHunger

■ Prudent Financial Projections

- **Year 1 (WFP Pilot Phase):** USD 75,000 budget fully covered by the WFP equity-free grant. Commercial revenue: USD 0. 100% focus on on-the-ground technical validation.
- **Year 2 (Initial Commercial Validation):** Moderate projected revenue of USD 15,000 - 25,000 (initial municipal technical support agreements and cooperative pilots).
- **Year 3 (Gradual Scaling):** Projected revenue of USD 45,000 - 60,000 expanding into additional vulnerable valleys and agricultural cooperatives.

■ Strategic Partnership with WFP

- **From Reaction to Anticipation:** Direct alignment with the World Food Programme's anticipatory action protocols.
- **Progressive Scaling:** From pilot basins to national coverage across Peru's 1,891 districts and vulnerable corridors in the Dominican Republic and El Salvador.
- **#DisruptHunger:** Transforming deterministic AI into a real-world shield to protect smallholder crops, livestock, and eliminate hunger.